TEXORA · LECTURE SHEET

Regenerated Fibres

Cellulose dissolved and born again as fibre: the viscose process step by step with its chemistry, the cleaner lyocell solvent route, modal and cuprammonium variants, and cellulose acetate — with the viscose–lyocell comparison the syllabus demands.

TE-2108 · MAN-MADE TEXTILE FIBERS · TOPIC 2 · LEVEL 2 TERM 1

1. The common idea

Cellulose — the polymer of cotton and wood — cannot be melted and does not dissolve in ordinary solvents, because its chains are locked together by hydrogen bonds (TE-2107). Regenerated-fibre technology therefore takes a detour: convert cellulose temporarily into a soluble form, extrude the solution through a spinneret, then convert it back to solid cellulose inside or after the jet. The starting cellulose is **dissolving pulp**: wood pulp purified to about 90–96 % alpha-cellulose, supplied as sheets. The final fibre is chemically cellulose again (except acetate, §5), but physically new — lower DP than the pulp, a shaped cross-section, and a skin–core structure set by the spinning bath.

2. The viscose process

2.1 Steeping and pressing

Pulp sheets are steeped in about 18 % sodium hydroxide solution at room temperature for roughly an hour. Cellulose swells and converts to **alkali cellulose**:

$$\text{Cell–OH} + \text{NaOH} \rightarrow \text{Cell–O}^{-}\text{Na}^{+} + \text{H}_2\text{O}$$

Excess liquor is pressed out, leaving a damp crumbly mass of alkali cellulose containing roughly equal weights of cellulose and alkali solution.

2.2 Shredding and ageing

The pressed sheets are shredded into crumbs to expose surface area, then **aged** in air at controlled temperature for some hours. Atmospheric oxygen, catalysed by the alkali, cuts the chains: DP falls from the pulp's roughly 700–900 to the 300–450 wanted in the fibre. Ageing is thus deliberate, controlled degradation (TE-2107 Topic 6) used as a process tool — under-aged dope is too viscous to spin; over-aged dope gives weak fibre.

2.3 Xanthation

The crumbs are tumbled with **carbon disulfide (CS_2)** — about 30 % on cellulose weight — for 2–3 hours. Hydroxyl groups react to form **cellulose xanthate**, an orange, soluble derivative (roughly one xanthate group per two glucose units):

$$\text{Cell–O}^{-}\text{Na}^{+} + CS_2 \rightarrow \text{Cell–O–CS–SNa (sodium cellulose xanthate)}$$

2.4 Dissolving, ripening, filtration

The xanthate crumbs dissolve in dilute NaOH to a viscous, honey-like solution — the **viscose** itself, typically 7–9 % cellulose. The dope is filtered (cloth/plate filters) to remove undissolved fibres that would block spinneret holes, deaerated under vacuum, and **ripened** for some days at controlled temperature. During ripening, xanthate groups redistribute and partially hydrolyse; viscosity first falls, then begins to rise as the solution approaches the point of coagulation — the spinner must catch the dope inside this window.

2.5 Spinning and regeneration

The dope is metered through spinnerets (thousands of holes for tow, dozens for yarn) into a **spin bath** of sulfuric acid (about 7–12 %), sodium sulfate and a little zinc sulfate, at about 45–55 °C. Two things happen in sequence:

1. **Coagulation** — the NaOH is neutralised, the sodium sulfate salts the xanthate out, and the jet solidifies as a gel filament (zinc forms a more resistant zinc xanthate skin, delaying the next step so the filament can be stretched first).
2. **Regeneration** — the acid decomposes the xanthate and cellulose re-forms:

$$2\ \text{Cell–O–CS–SNa} + H_2SO_4 \rightarrow 2\ \text{Cell–OH} + 2\ CS_2\uparrow + Na_2SO_4$$

The filaments are stretched between rollers of increasing speed (orientation, crystallinity and strength rise), then washed free of acid and sulfur, desulfurised, bleached, given a softening finish, dried, and conditioned. Cross-section: a **serrated, irregular outline** with a distinct skin — the spinneret hole is round, but the skin forms first and the core collapses inward as it regenerates, wrinkling the outline.



then: stretching → washing → desulfurising → bleaching → finish → drying (filtering + ripening of the dope occur between dissolving and spinning)

CS_2 AND H_2S EMISSIONS MAKE THIS THE POLLUTING ROUTE — RECOVERY SYSTEMS ARE ESSENTIAL

Figure 1 — The viscose process flow. Pulp becomes alkali cellulose, is aged, xanthated, dissolved, ripened and wet-spun back into cellulose in an acid bath.

3. Properties of viscose

PROPERTY	VISCOSE BEHAVIOUR
Tenacity (dry)	Moderate, ≈ 2–2.5 g/den typical for regular staple
Wet strength	Falls to roughly 50–60 % of dry — the family's weak point; water breaks the hydrogen bonds between its low-DP, largely amorphous chains
Moisture regain	High, ≈ 11–13 % — comfortable, absorbent, dyes easily with cotton dyes
Elongation	High (≈ 15–25 %), low elastic recovery — garments bag and crease
Behaviour to heat/alkali	Chars like cotton; tendered by strong alkali and oxidising bleach if uncontrolled

WORKED EXAMPLE — WET-STRENGTH LOSS IN NUMBERS

A regular viscose filament yarn of 150 denier has dry tenacity 2.2 g/den and wet tenacity 1.1 g/den.

Dry breaking load = 150 × 2.2 = **330 g**; wet breaking load = 150 × 1.1 = **165 g** — exactly **50 % retained**.

A comparable cotton yarn retains about 105–110 % of its dry strength when wet (cotton is stronger wet), and polyester retains 100 %. This single comparison explains why regular viscose is kept out of high-stress wet applications and why the improved variants of §4 and §6 were developed.

4. Modal and high-wet-modulus (HWM) fibres

Modal is viscose chemistry modified for wet strength: higher-DP pulp, less ageing, modified bath additives and much higher stretch during spinning give more orientation and crystallinity. Result: wet strength retention and dimensional stability far above regular viscose, cotton-like hand, and durability

through repeated washing — modal is the standard regenerated fibre for underwear and blends with cotton. **Polynosic** fibres are a related high-performance branch of the same idea.

5. Lyocell — the direct-solvent route

Lyocell dissolves pulp **directly**, with no chemical derivative: the solvent is **N-methylmorpholine-N-oxide (NMMO)** in a little water, which breaks cellulose's hydrogen bonds physically at about 90–120 °C to give a clear dope. The dope is dry-jet wet spun: jets cross a short air gap (where they are stretched and oriented) into a dilute NMMO water bath that washes the solvent out and precipitates the cellulose. Consequences:

- **Closed loop** — over 99 % of the NMMO is recovered and reused; no CS_2, no xanthate, minimal effluent: the environmental flagship of regenerated fibres.
- **Higher DP retained** (no ageing step) and strong orientation from the air-gap draw → tenacity well above viscose, dry and especially wet (wet retention ≈ 80–85 %).
- **Round, smooth cross-section** and a tendency to **fibrillate** — surface micro-fibrils peel under wet abrasion, a defect for smooth fabrics but the source of peach-skin effects when controlled by enzyme treatment.

FEATURE	VISCOSE	LYOCELL
Solvent system	Chemical route: NaOH + CS_2 (xanthate derivative)	Direct physical solvent: NMMO/water
Regeneration	Acid bath decomposes xanthate	Water bath extracts NMMO
Solvent recovery	Partial; CS_2/H_2S emissions are a major burden	> 99 % closed loop
DP of fibre cellulose	Reduced by ageing to ≈ 300–450	Kept close to pulp DP
Cross-section	Serrated, skin–core	Round, uniform
Wet strength retention	≈ 50–60 %	≈ 80–85 %
Special behaviour	Soft drape, cheap, huge scale	Fibrillation; premium price

6. Cuprammonium and acetate

Cuprammonium rayon (cupro) dissolves cellulose in an ammoniacal copper solution (Schweizer's reagent, $[Cu(NH_3)_4](OH)_2$) and regenerates it in dilute acid/water. It gives exceptionally fine, silky filaments for linings, but copper/ammonia recovery costs have shrunk it to a niche.

Cellulose acetate is regenerated *and modified*: cellulose is acetylated with acetic anhydride to the triacetate, partially hydrolysed back to the **secondary (di-)acetate** (about 2.4 of the 3 hydroxyls per unit acetylated), dissolved in acetone and **dry-spun** (Topic 5 of this course treats spinning methods fully). Acetate is *not* cellulose: it is thermoplastic (softens/melts with a hot iron's carelessness), has low regain (≈ 6 %), dyes with disperse dyes, and gives silk-like drape for linings and evening fabrics. **Triacetate** (nearly fully acetylated) is more heat-resistant and holds pleats.

Glossary

Dissolving pulp — purified wood pulp of high alpha-cellulose content used for regenerated fibres. **Alkali cellulose** — the swollen sodium-cellulosate formed in steeping. **Xanthate** — the soluble orange cellulose derivative formed with CS_2. **Ripening** — controlled ageing of the viscose dope to the spinnable state. **NMMO** — N-methylmorpholine-N-oxide, lyocell's direct solvent. **Fibrillation** — splitting of surface micro-fibrils from a lyocell filament under wet abrasion. **HWM** — high wet modulus: modal-class fibres engineered for wet strength.

Key points

• Viscose: steep → press → shred → age (DP down) → xanthate with CS_2 → dissolve → ripen → wet-spin into acid, where the xanthate is decomposed and cellulose regenerated.
• Regular viscose keeps only about half its strength when wet — the direct result of low DP and low crystallinity.
• Modal fixes wet strength within viscose chemistry; lyocell abandons the chemistry for a direct solvent (NMMO) in a > 99 % closed loop, keeping DP and wet strength high.
• Acetate is a different material, not cellulose: acetone-soluble, thermoplastic, disperse-dyed.
• Environmental ranking of the family follows the solvent system: lyocell cleanest, viscose burdened by CS_2.

Practice

1. Put the viscose stages in order and give the chemical purpose of each: dissolving, steeping, spinning/regeneration, ageing, xanthation.
2. A spinner wants stronger viscose filament. Which two process changes raise orientation and crystallinity, and where in the sequence do they act?
3. Explain, from chain structure, why viscose loses half its strength when wet while cotton gains strength.
4. Lyocell is called the "closed-loop" fibre. What exactly is looped, and what viscose emission does this eliminate?

5. An "acetate" lining fabric is scorched by a hot iron while a viscose one only chars at higher heat. What structural difference explains the scorching?

Answer key

1. Steeping (form alkali cellulose, swell the pulp) → ageing (oxidative DP reduction to spinnable viscosity) → xanthation (make cellulose soluble as xanthate) → dissolving (make the viscose dope in dilute alkali) → spinning/regeneration (acid decomposes xanthate, cellulose re-forms as filament).
2. (i) Higher stretch between the godet rollers immediately after the bath — draws chains parallel before the structure sets. (ii) Slower regeneration (zinc-modified bath / lower acid action) so stretching happens while the filament is still a deformable gel rather than a set solid.
3. Viscose chains are short (DP ≈ 300–450) and the fibre is mostly amorphous with a weak skin–core arrangement, so water penetrates, replaces inter-chain hydrogen bonds and plasticises the load-bearing structure. Cotton's chains are far longer and highly crystalline; water enters only the amorphous fraction, and the swollen chains pack and grip better when wet than dry.
4. The solvent NMMO is washed out in the spin bath, recovered from the dilute wash water by evaporation, and returned to dissolve more pulp — over 99 % circulates. This eliminates the CS_2 (and its H_2S by-product) emissions inherent in making and regenerating cellulose xanthate.
5. Acetate is cellulose ester, a thermoplastic with a softening/melting range reachable by a hot iron, so it glazes and fuses (scorches) rather than merely charring. Viscose is pure cellulose, which has no melting point and degrades only at higher temperatures.
